

day 26 | 251029 W

```
In [1]: %matplotlib ipynb
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import numpy as np
import matplotlib.pyplot as plt
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In [ ]: # define the differential equation
def f(r, t): # don't forget r is a vector!
    x = r[0]
    y = r[1]
    w = 1.0
    fx = x*y-x # this is the dx/dt equation!
    fy = y - x*y + np.sin(w*t)**2 # this is the dy/dt equation!
    return(np.array([fx, fy]))

# define the boundary condition
a = 0.0
b = 10.0
N = 1000
dt = (b-a)/N

# Runge-Kutta 4th order

r = np.array([1.0, 1.0]) # initial condition

tpoints = np.arange(a, b, dt)

xpoints = []
ypoints = []

for i in tpoints:
    xpoints.append(r[0])
    ypoints.append(r[1])
    k1 = dt*f(r, i)
    k2 = dt*f(r+1/2*k1, i+1/2*dt)
    k3 = dt*f(r+1/2*k2, i+1/2*dt)
    k4 = dt*f(r+k3, i+dt)
    r = r + (k1+2*k2+2*k3+k4)/6
```